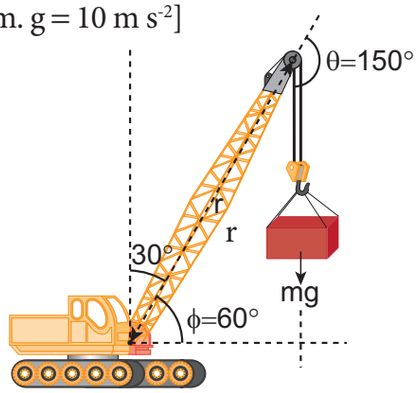
EXAMPLE 5.9

A crane has an arm length of 20 m inclined at 30° with the vertical. It carries a container of mass of 2 ton suspended from the top end of the arm. Find the torque produced by the gravitational force on the container about the point where the arm is fixed to the crane. [Given: 1 ton = 1000 kg; neglect the weight of the arm. g = 10 m s^{-2}]



Solution

Note

In many problems, the angle θ between $\vec{r}$ and $\vec{F}$ will not be directly given. Thus, the students must get accustomed to identify and denote always the angle between the $\vec{r}$ and $\vec{F}$ as θ. The other angles in the arrangement may be denoted as α, β, ϕ etc.

The force F at the point of suspension is due to the weight of the hanging mass.

$$F = mg = 2 \times 1000 \times 10 = 20000 \text{ N};$$
The arm length, r = 20 m

We can solve this problem by three different methods.

Method – I

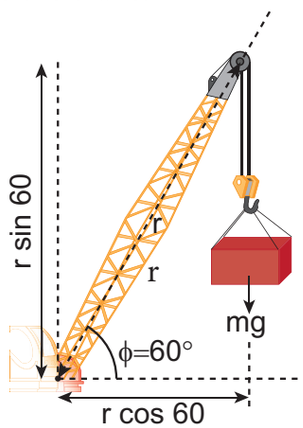
The angle (θ) between the arm length (r) and the force (F) is, θ = 150°

The torque (τ) about the fixed point of the arm is,

$$\tau = r F \sin\theta$$
$$\tau = 20 \times 20000 \times \sin\left(150^\circ\right)$$
$$= 400000 \times \sin\left(90^\circ + 60^\circ\right)$$
$$[\text{here, } \sin\left(90^\circ + \theta\right) = \cos\theta]$$
$$= 400000 \times \cos\left(60^\circ\right)$$
$$= 400000 \times \frac{1}{2} \quad \left[\cos 60^\circ = \frac{1}{2}\right]$$
$$= 200000 \text{ N m}$$
$$\tau = 2 \times 10^5 \text{ N m}$$

Method – II

Let us take the force and perpendicular distance from the point where the arm is fixed to the crane.



$$\tau = \left(r \perp\right) F$$
$$\tau = r \cos\phi \, mg$$
$$\tau = 20 \times \cos 60^\circ \times 20000$$
$$= 20 \times \frac{1}{2} \times 20000$$
$$= 200000 \text{ Nm}$$
$$\tau = 2 \times 10^5 \text{ Nm}$$

Method – III

Let us take the distance from the fixed point and perpendicular force.

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